TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Agents for Autonomous Decision-Making**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication (), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software Solution / System Integration	R&D / Engineering Services	Industrial Facility Survey / Design / Forecasting / Evaluation / Management Technologies
Concept of Proposed Technology	Nowadays, **factories operate in complex, tightly coupled networks of production lines**, suppliers, logistics partners, and customer commitments. As a result, optimization has shifted from **machine-performance focus to a multi-variable problem** that must balance throughput with delivery reliability, maintenance cost, energy usage, quality risk, and work-in-progress ([7], [10], [13]). Managers, team-leads, and **engineers cannot realistically track all constraints**, trade-offs, and fast-changing signals, nor implement the right decisions quickly enough by hand [11]. The central challenge is therefore to **decide and act while disruptions are still manageable** and before deviations propagate downstream [8]. The proposed technology **"Agents for Autonomous Decision Making"** aims to make operational decision making autonomous: it continuously detects deviations between plan and reality and selects corrective actions under multiple constraints. A team of **specialized agents coordinates**			

	the decision process across reliability, process optimization, quality, compliance, and tactical planning. For example, maintenance scheduling with spare-part readiness, bottleneck-driven reallocation of resources, and quality-drift corrections. The **technology closes the loop by feeding validated deviations and decisions** back into the planning system to improve future plans and reduce recurring variability. It includes auditable decision traces and **human-in-the-loop approval gates** when uncertainty rises or potential impact reaches a threshold, to ensure safe and accountable autonomy. Typical uses include autonomous rescheduling during disruptions, proactive maintenance planning, and coordinated responses to quality or supply issues.
Rationale for Joint R&D Project	Factories must continuously improve their operations to remain competitive, but operational control has become a multi-variable decision problem: throughput, delivery reliability, maintenance cost, energy use, quality risk, and work-in-progress must be balanced under time pressure. The technical and operational risks are high, because decisions must be made under uncertainty, within strict constraints, and with clear accountability. From the Korean SME's perspective, pursuing this alone is challenging for four reasons. First, high-quality data from machines and shopfloor/business systems must be available. In practice, data quality and integration are the main bottlenecks. TNO has extensive experience integrating heterogeneous equipment into (generic) IoT platforms [3] and applying Industry 4.0 concepts and standards ([1], [4], [5], [15]), creating reusable digital foundations that can also accelerate other initiatives beyond this project. Second, AI specialists with experience in manufacturing are needed to select and implement the right kind of reasoning algorithms. TNO has a proven track record in applied manufacturing innovation, translating industrial challenges into working solutions in projects ([2], [6]), and advancing factory autonomy in programs such as [13]. Third, GenAI evolves rapidly and must be engineered carefully to avoid failure modes such as hallucinations that result in the wrong course of action [12]. In [14], we developed and tested an LLM-based control loop to support operators in a one-piece-flow production setting. Finally, successful adoption requires alignment with long-term strategy, business value, and change management ([7], [10]). With TNO, the Korean SME gains a partner that accelerates data readiness, builds a working prototype fast, and makes sure the solution is safe, reliable, and useful for daily operations. From TNO's perspective, developing robust and generalizable methods for autonomous decision-making requires testing and validation in real industrial environments where constraints, disturbance patterns, planning interfaces, and data imperfections are present by default. Prior research on human-centered, agent-based manufacturing and structured digital asset representations ([4], [5], [9]) provides a methodological basis, but practical verification requires an industrial partner and daily-use feedback loops. SMEs are an ideal partner because their scale enables rapid iteration, direct access to domain experts, and practical validation cycles, while still representing the real-world complexity of manufacturing operations. Collaboration is necessary because neither party can achieve the objective alone: TNO can de-risk and structure development using proven Industry 4.0 methods and state-of-the-art GenAI techniques, while the SME provides the operational context, data access, and iteration speed needed to test, refine, and validate autonomous decision-making in practice. Joint execution ensures that the system is not only technically sound but also operationally adoptable. Ultimately, the goal is to turn state of the art research into a solution that the SME can run in daily operations and build upon over time as part of a longer-term trajectory toward increased factory autonomy.
Key Words	1. Autonomy 2. Decision-Making 3. Gen-AI 4. Multi-Agent 5. Industry 4.0

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	TRL 3-4	End of the Project	TRL 6
R&D Objectives	**Agents for Autonomous Decision-Making** • **Multi-modal data foundation for autonomous decisions**: develop a pipeline to collect, clean, align, and persist in event, planning, quality, maintenance, and logistics data in a database optimized for agentic decision-making. • **Time-series "memory" generation for GenAI**: implement methods that convert continuous signals into compact, queryable memory objects (states, change-points, etc.) that preserve traceability to the original data. • **Disturbance modelling and detection**: develop technology to recognize and characterize disturbances (deviations from expected scenarios) and to quantify their impact on objectives and constraints. • **Multi-level reasoning for decision quality**: develop reasoning methods that combine situational understanding, constraint checking, trade-off analysis, and counterfactual/what-if evaluation to select effective actions under uncertainty. • **Multi-agent operational decision-making**: build specialized agents (depending on the SME needs), e.g. for coordinated decisions across maintenance/reliability, throughput and resource allocation, quality stabilization, and tactical replanning. • **Closed-loop control architecture including an LLM**: Design and implement a safe control loop using state-of-the-art Object-Reflect-Plan-Act (ORPA) or Observe-Orient-Decide-Act (OODA) technology. • **Decision traceability and accountability**: implement decision traces that capture inputs, constraints, rationale, alternatives considered, and expected effects, producing auditable records suitable for technical and managerial review. • **GenAI-ready processing of engineering artefacts**: develop ingestion and interpretation of CAD/CAM and related engineering files so agents can use product and process intent (geometry, tolerances, routings) as decision context. • **Human-in-the-Loop interfaces for oversight, validation and refinement**: develop an interface and workflow for approval gates, exception handling, and iterative refinement, ensuring operators and engineers can intervene, steer behavior, and override decisions when needed.			
R&D Target Specifications	**Disturbance detection performance (pilot scope)** • Metric: Percentage of disturbances correctly identified from a predefined and labelled set • Target: ≥ 75% precision on the evaluation set **Decision options generated per disturbance (action coverage)** • Metric: Average number of decision options generated per detected disturbance, including at least one feasible option that satisfies constraints. • Target: ≥ 2 candidate decisions per disturbance **Human acceptance of recommended decisions (effectiveness proxy)** • Metric: Percentage of cases where an SME expert rates the top recommendation as "reasonable and usable," even if not optimal. • Target: ≥ 50% of top recommendations rated usable; ≤ 10% rated "clearly wrong for the situation." **Decision trace completeness (accountability)** • Metric: Share of decisions that include a trace with (a) triggering disturbance, (b) referenced inputs, (c) constraints checked, (d) rationale, and (e) expected impact. • Target: ≥ 80% of decisions include a complete trace			

	Supervised autonomy runtime (progressive autonomy) • Metric: Continuous time in which the system runs in supervised autonomous mode (auto-generate decisions; execution requires approval or predefined safe rules). • Target: Demonstrate 30 minutes continuous operation, then 60 minutes, in a controlled pilot; 0 safety-rule violations during these periods. Note, the feasibility of the presented targets depends strongly on the SME's willingness to experiment in operations and on the maturity of the available IT infrastructure (data access, system integration, latency, security constraints, and change-management capacity). For that reason, the quantitative targets must be refined at project start based on the agreed pilot scope, accessible data sources and APIs, and the level of operational experimentation permitted.
R&D Plan	**Research approach** The project combines Industry 4.0 methods (UNS, IoT, ISA-95, ISO-22400) with GenAI patterns (tool calling, structured outputs, human-in-the-loop) to build and validate an autonomous decision-making loop for factory operations. The work follows an iterative engineering-research cycle: (1) formalize processes, roles, constraints, and KPIs; (2) structure and connect operational data early via an SME-specific Unified Namespace (UNS) and IoT connectivity; (3) develop and evaluate time-series-to-"memory" algorithms that convert continuous signals into GenAI-ready decision context; (4) implement a decision-making stack (agents + reasoning + constraint checks + decision traces) that can propose and, under controlled conditions, execute actions; and (5) validate staged autonomy in the factory environment with human oversight and measurable performance targets. Process descriptions and KPI definitions are aligned with ISA-95 and ISO-22400 and captured using BPMN. From early in the project, a working prototype is maintained and expanded in short iterations, so integration constraints and operational feedback shape the technical choices. Throughout the project, regular interactive working sessions with the SME steer disturbance selection, autonomy boundaries, algorithm choices, and interface design. Multiple milestones are designed to deliver usable increments that can be tested in realistic operations and refined based on measurable outcomes and user review. **Phases, key tasks, methodologies, milestones, and timeline** The project execution is split into six distinct phases, each of which can take between 3 and 6 months, depending on the concrete use case at hand, and the development results. In the situation when phase 6 is reached prior to the end of the project duration, an extra phase will be introduced, focused on the introduction of more complex functionalities to the system or expansion of the processes. **Phase 1: Baseline, constraints, and disturbance catalogue** Key tasks • Inventory factory processes, organization, and KPIs. • Inventory software packages, data flows, and available interfaces/APIs. • Define the initial "disturbance catalogue" (deviations from expected scenarios) and decision scope for the pilot (what the system may recommend vs. what it may execute) together with the SME. • Discuss governance: approval gates, safety/compliance boundaries, security and decision-trace requirements. Methodologies • BPMN for processes; ISA-95 mapping; ISO-22400 KPIs. Milestones • M1: Process maps + KPI dictionary + disturbance catalogue v1.

- M2: Software/data-flow gap analysis + pilot scope and autonomy boundaries.

Phase 2: UNS build-out and data readiness for control-loop use

Key tasks

- Design the SME-specific UNS, following UNS design recommendations and i4.0 principles. Using the company-chosen IoT platform (open source or commercial).
- Connect the first set of operationally critical data sources to the UNS (time-series and events).
- Apply the TNO UNS MCP server to the UNS and expose tools that can query the raw data signals.
- Identify and collect supporting information sources (manuals, SOPs, schematics) for later grounding and operational context.

Methodologies

- UNS design patterns; IoT integration; data quality checks.

Milestones

- M3: UNS v1 with initial connected sources and validated tags/events.

Phase 3: Time-series -> "memory" algorithms for disturbance handling

Key tasks

- State-of-the-art review of time-series representation and summarization approaches for LLM consumption.
- Create a benchmark dataset from the SME environment (representative lines/assets)
- Implement a memory pipeline that produces structured memory objects consumable by the decision agents.
- Evaluate prototype with benchmark dataset

Methodologies

- Time-series segmentation, summarization, and anomaly/state-change detection; structured memory schemas; expert review against labelled disturbances.

Milestones

- M4: Memory pipeline v1 producing traceable, queryable memory objects for the pilot disturbance classes.

Phase 4: Decision-making architecture and decision traces (supervised autonomy)

Key tasks

- Design the decision-making architecture based on a state-of-the-art review combined with requirements from the SME. Current SotA research suggests this will be a multi-agent GenAI architecture.
- Implement the decision-making architecture, which most likely will consist of specialized agents + reasoning engine + constraint checks for candidate action generation and ranking.
- Prompt engineering for decision traces (inputs, constraints checked, alternatives, rationale, expected impact) and establish review workflows with decision makers.
- Begin closed-loop integration with the planning system (what can be updated, how frequently, and under which approvals).

Methodologies

- Tool calling; structured outputs; multi-step reasoning; constraint/guardrail design;

traceability and audit patterns.

Milestones

- M5: Prototype v1 capable of generating disturbance-linked decisions with complete decision traces under human approval.

Phase 5: Human-in-the-Loop interfaces and document grounding

Key tasks

- Agree on preferred interaction approach and build the factory-specific HMI for decision review, validation, and override.
- Extend the system with RAG over manuals/SOPs/schematics to support operational justification, procedures, and constraints during decision formulation.

Methodologies

- Human-in-the-loop design; prompt engineering; RAG pipeline design and tuning; usability testing with operators/engineers/managers.

Milestones

- M6: Prototype v2 with decision-review interface, approval gates, and document-grounded decision support.

Phase 6: Staged autonomy pilot and planning feedback loop validation

Key tasks

- Pilot staged autonomy on selected disturbance classes: first recommend-only, then supervised execution within defined safe boundaries.
- Measure project R&D target specifications (disturbance detection coverage, decision options per disturbance, decision-trace completeness, response time, and continuous supervised autonomy runtime).
- Validate closed-loop feedback to the planning system (updating parameters, rescheduling recommendations, and recording outcomes) and document operational learnings.
- Technology transfer to SME

Methodologies

- Field validation; structured evaluation sessions; KPI and constraint monitoring; post-mortem analysis of disturbances and decisions.

Milestones

- M7: Final architecture + report against targets + rollout roadmap and scale-up plan.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• **Apply Industry 4.0 principles in the SME context**: translate ISA-95 and ISO-22400 concepts into a practical, operational, SME-specific data and KPI framework. Guide the setup of an SME-specific UNS. • **Provide UNS expertise and assets**: deliver and tailor TNO UNS guidelines (naming conventions, data semantics, integration patterns, etc) to ensure consistent and scalable data structuring for real-time decision-making use. • **Support IoT connectivity and data acquisition**: advise on IoT architectures and interfaces to extract, contextualize, and securely expose machine and shopfloor data, including practical integration options and constraints. • **Support process formalization where needed**: consult the SME on documenting processes and responsibilities (e.g. BPMN-style diagrams) to link data, KPIs, and operational workflows. Consult the SME to define decision constraints, approval rules, and operational boundaries required for controlled autonomy. • **Conduct state-of-the-art research and selection**: perform and document the SoTA analysis for time-series-to-"memory" representations and decision-oriented reasoning methods. Define evaluation criteria and recommend the most suitable algorithms/approaches for the pilot scope. • **Develop the autonomous decision-making proof-of-concept**: implement a POC that detects disturbances, generates decision options under constraints, produces auditable decision traces, and supports staged autonomy with human approval gates.
Expected R&R of Korean SME Partner	• **Define operational objectives and business KPIs**: clarify the SME's priority KPIs and decision objectives (e.g., delivery reliability, maintenance cost, energy, WIP, quality risk), including calculation rules, target ranges, and accepted "ground truth." • **Provide process, planning, and constraint knowledge**: share detailed information on shopfloor and planning processes and collaborate on formalized descriptions (roles, handovers, decision points), including explicit operational constraints and rules (capacity, sequencing constraints, maintenance windows, safety limits). • **Enable data and system integration**: provide access to factory data and planning/operations system interfaces via APIs (machines, MES/ERP, quality, maintenance, scheduling/planning), and support data governance, permissions, and cybersecurity requirements. • **Contribute connectivity for non-integrated assets**: contribute technical resources to connect machines or data sources that are not yet reachable via IoT, including troubleshooting and onboarding of legacy equipment. • **Co-develop integration at SME**: allocate technical staff to co-develop integration components, configuration, and deployment elements, using the SME's IT infrastructure and engineering practices. • **Supply disturbance cases and decision scenarios**: provide prioritized disturbance examples, historical incidents, and representative decision scenarios to drive development, benchmarking, and evaluation of decision quality and constraints handling. • **Pilot operation, evaluation, and feedback**: run the tooling in supervised operational trials, evaluate recommendations and autonomous behavior against real constraints, and deliver structured feedback (accept/reject reasons, missing context, required safeguards).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will enable faster and more consistent operational responses to disturbances by detecting deviations between plan and reality, selecting corrective actions under constraints, and closing the loop by updating the planning system. It will reduce decision latency and coordination overhead during disruptions, so issues are addressed while they are still manageable and before they propagate downstream. It will also establish a practical governance approach for controlled autonomy, including approval gates and auditable decision traces. **Technical outcome**: Working prototype of a multi-agent-based decision-making system that coordinates e.g. reliability, process optimization, quality, compliance, and tactical planning decisions. Key technical results include a validated time-series-to-"memory" method for disturbance characterization, a reasoning and constraint-checking engine for generating and ranking decisions, and a closed-loop integration that feeds validated deviations and decisions back into both the decision-making system and the planning environment. The solution will produce decision traces that document triggers, inputs, constraints, alternatives, rationale, and expected impact. **Commercialization potential**: The technology can be productized as an "autonomous operations" software layer that recommends and, in supervised modes, executes decisions such as rescheduling, maintenance window planning, bottleneck mitigation, and quality stabilization. Differentiation comes from multi-agent coordination across operational domains, explicit constraint handling, and built-in accountability through decision traces and human oversight controls. **Economic potential:** Operational optimization has become a multi-variable decision problem where throughput, delivery reliability, maintenance cost, energy usage, quality risk, and work-in-progress jointly determine financial outcomes. Human decision makers cannot continuously track all constraints, trade-offs, and fast-changing signals, especially during disruptions. By making timely, constraint-aware decisions and acting before deviations cascade, the proposed technology reduces avoidable losses, improves plan stability, and increases the effectiveness of resources and capital. In addition, it offloads routine day-to-day coordination and decision work (e.g., frequent replanning, disturbance triage, and constraint checks), freeing managers, team-leads, and engineers to focus on higher-impact decisions that require human judgment, creativity, and strategic trade-off setting. The result is lower operational cost and improved overall factory performance through faster, better-aligned decisions.

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